# Analyse III

# TD n° 1: Intégrales multiples

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2e Année CPST 2020/2021

# Plan


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## Exercice 1

*Calculer les intégrales doubles suivantes :*

a). $\iint_D xydxdy$; $D = \{(x,y) \in \mathbb{R}^2;\ x \geq 0;\ y \geq 0;\ x+y \leq 1\}$

b). $\iint_D |xy|\, dxdy$; $D = \left\{(x,y) \in \mathbb{R}^2;\ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\right\}$

c). $\iint_D (x+y)dxdy$; $D =$*intérieur du triangle de sommets*

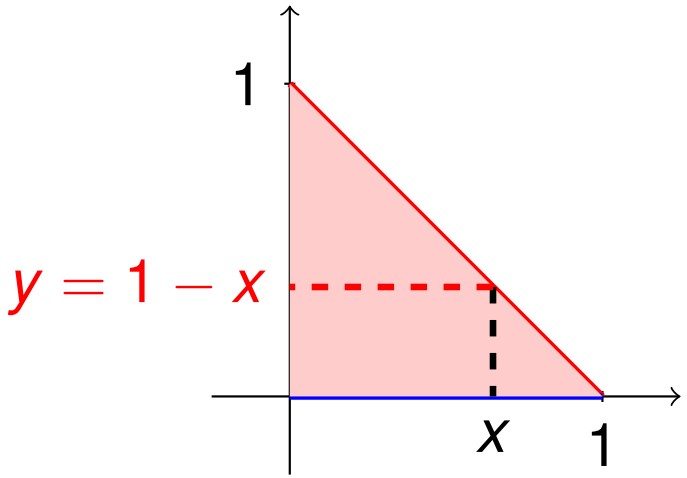
$$A(0,0),\quad B(1,1),\quad C(2,0).$$

d). $\iint_D \frac{xy}{(x^2+y^2)^2} dxdy$;

$$D = \left\{(x,y) \in \mathbb{R}^2;\ x^2+y^2 \leq 1,\ x+y \geq 1\right\}.$$

***Solution :***

**a)** $\iint_D xydxdy$; $D = \{(x,y) \in \mathbb{R}^2;\ x \geq 0;\ y \geq 0;\ x+y \leq 1\}$

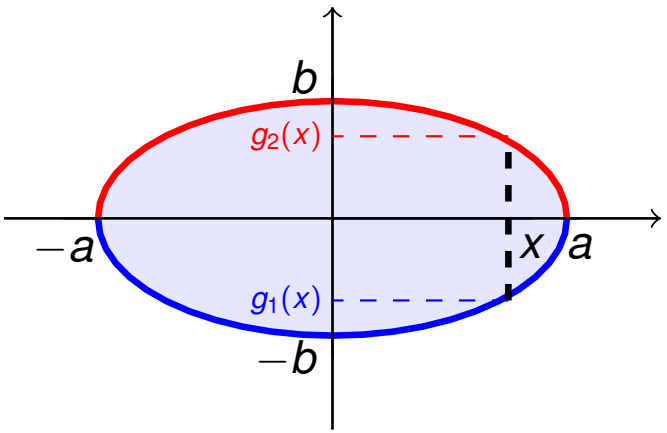



$$I = \int_0^1 \left( \int_0^{1-x} xydy \right) dx = \frac{1}{24}.$$

**b)** $\iint_D |xy|\,dxdy$; $D = \left\{(x,y) \in \mathbb{R}^2;\ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\right\}$.

On peut décrire $D$ sous la forme

$$D = \left\{(x,y) \in \mathbb{R}^2;\ -a \leq x \leq a,\ g_1(x) \leq y \leq g_2(x)\right\}$$



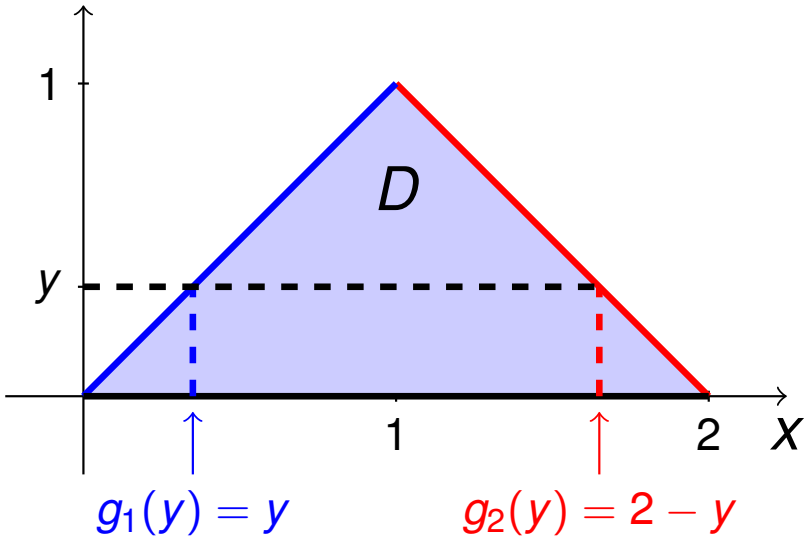

où $g_2(x) = -g_1(x) = \frac{b}{a}\sqrt{a^2 - x^2}$.

En appliquant le théorème de Fubini on a

$$\begin{aligned}\iint_D |xy|dxdy &= \int_{-a}^{a} \left( \int_{-g_2(x)}^{g_2(x)} |xy|dy \right) dx \\ &= 4 \int_0^a \left( \int_0^{g_2(x)} |xy|dy \right) dx \\ &= \frac{a^2 b^2}{2}.\end{aligned}$$

**c)** $I = \iint_D (x+y)dxdy$ où $D$ est l'interieur du triangle de sommets $A = O = (0,0)$, $B = (2,0)$ et $C = (1,1)$.

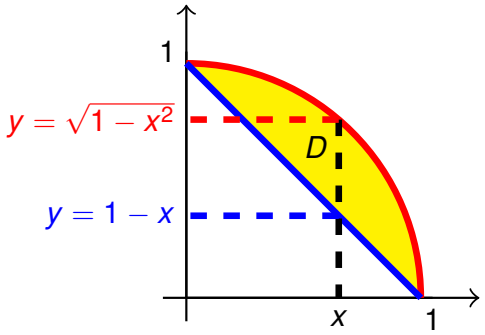
On a $D = \{(x,y) \in \mathbb{R}^2;\ 0 \leq y \leq 1,\ y \leq x \leq 2-y\}$




$$I = \int_0^1 \left( \int_y^{2-y} (x+y)dx \right) dxy = \int_0^1 (2-2y^2)dy = \frac{4}{3}.$$

**d)** $\iint_D \frac{xy}{(x^2+y^2)^2}\,dxdy$;

$$D = \left\{(x,y) \in \mathbb{R}^2;\ \ x^2+y^2 \le 1,\ \ x+y \ge 1\right\}.$$




$$I = \int_0^1 \left( \int_{1-x}^{\sqrt{1-x^2}} \frac{xy}{(x^2+y^2)^2}\,dy \right) dx = \int_0^1 \left( \frac{x}{2(2x^2-2x+1)} - \frac{x}{2} \right) dx$$

$$= \int_0^1 \left( \frac{4x-2}{8(2x^2-2x+1)} + \frac{1}{4(2x^2-2x+1)} - \frac{x}{2} \right) dx$$

$$= \left[ \frac{1}{8}\ln(2x^2-2x+1) + \frac{1}{4}\operatorname{arctg}(2x-1) - \frac{x^2}{4} \right]_0^1 = \frac{\pi}{8} - \frac{1}{4}.$$

## Exercice 2

*Calculer* $I = \iint_D f(x,y)dxdy$ *en utilisant le changement de variables indiqué :*

a). $f(x,y) = \frac{1}{x^2+y^2+1}$; $\quad D: \ x^2 + y^2 \leq 1$;
*(coordonnées polaires).*

b). $f(x,y) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^{-1/2}$; $\quad D: \ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$
$(x = ar\cos\theta; \ y = br\sin\theta; \ \theta \in [0, 2\pi[\,; \ r \in ]0,1])$.

c). $f(x,y) = 1$; $D =$ *limité par les courbes d'équations :*

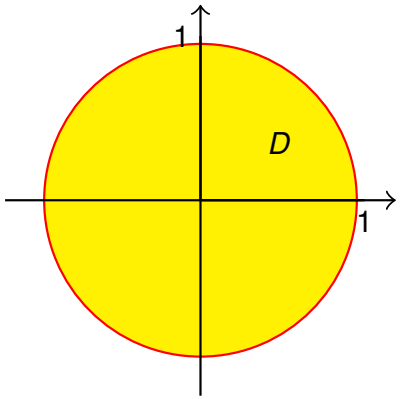
$$y = ax; \quad y = x/a; \quad y = b/x; \quad y = 1/bx;$$

$$(a > 1; b > 1; x > 0)\,.$$

*( changements de variables ;* $x = u/v; \quad y = uv$*).*

***Solution :***

**a)** $I = \iint_D \frac{1}{x^2+y^2+1} dxdy; \quad D: \ x^2 + y^2 \leq 1.$

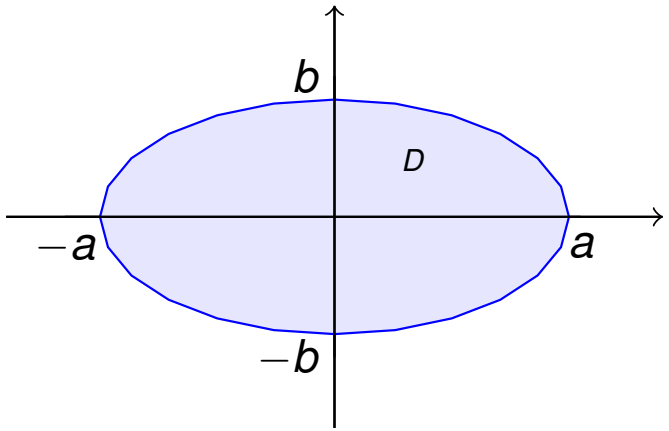



Posons $x = r\cos\theta; \ y = r\sin\theta, \quad \det(J(r,\theta)) = r.$

$$(x,y) \in D \Leftrightarrow (r,\theta) \in \Delta = ]0,1] \times [0,2\pi[.$$

$$I = \int_0^{2\pi} d\theta \int_0^1 \frac{r}{r^2+1} = \pi \ln 2.$$

**b)** $I = \iint_D \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^{-\frac{1}{2}} dxdy; \quad D : \ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1.$



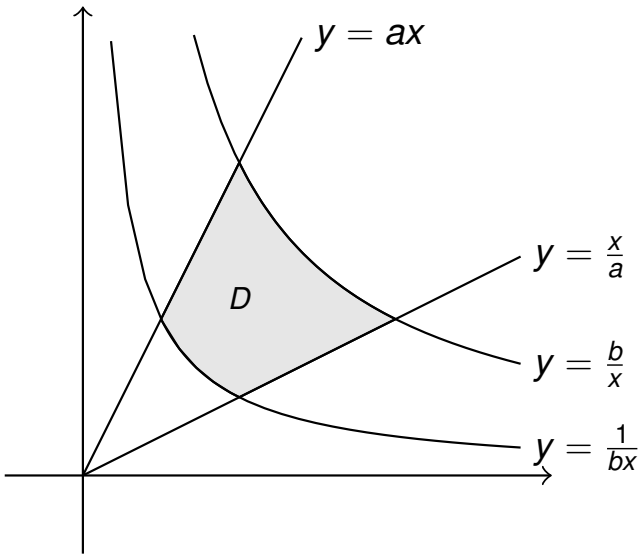

Posons $x = ar\cos\theta;\ y = br\sin\theta, \quad \det(J(r,\theta)) = abr.$

$$(x,y) \in D \Leftrightarrow (r,\theta) \in \Delta = ]0,1] \times [0,2\pi[.$$

$$I = \iint_{]0,1]\times[0,2\pi[} \frac{1}{r} ab.rdrd\theta = 2\pi ab.$$

**c)** $I = \iint_D dxdy$; $D : y = ax$; $y = x/a$; $y = b/x$; $y = 1/bx$;

$(a > 1; b > 1; x > 0)$.




Posons $x = \dfrac{u}{v}$, $y = uv$, alors

$$\begin{cases} u^2 = xy \\ v^2 = \dfrac{y}{x}. \end{cases} \Longleftrightarrow \begin{cases} u = \sqrt{xy} \\ v = \sqrt{\dfrac{y}{x}}. \end{cases}$$

$$
\begin{aligned}
(x,y) \in D &\iff \frac{x}{a} \leq y \leq ax, \quad \frac{1}{bx} \leq y \leq \frac{b}{x} \\
&\iff \frac{1}{a} \leq \frac{y}{x} \leq a, \quad \frac{1}{b} \leq xy \leq b \\
&\iff \frac{1}{a} \leq v^2 \leq a, \quad \frac{1}{b} \leq u^2 \leq b \\
&\iff (u,v) \in \Delta = \left[\frac{1}{\sqrt{b}}, \sqrt{b}\right] \times \left[\frac{1}{\sqrt{a}}, \sqrt{a}\right]
\end{aligned}
$$

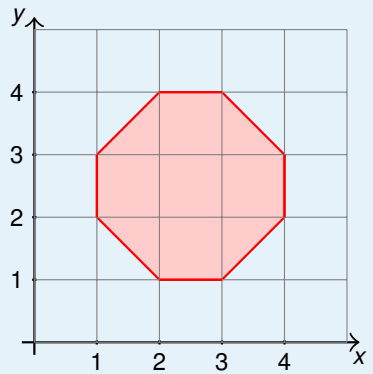
$$
\text{Posons} \quad (x,y) = \varphi(u,v), \quad \text{alors} \quad \det\left(J_\varphi(u,v)\right) = 2\frac{u}{v}.
$$

$$
I = \iint_\Delta 2\frac{u}{v} du dv = 2\left(\int_{\frac{1}{\sqrt{b}}}^{\sqrt{b}} u du\right)\left(\int_{\frac{1}{\sqrt{a}}}^{\sqrt{a}} \frac{1}{v} dv\right) = \left(b - \frac{1}{b}\right) \ln a.
$$

## Exercice 3

*Soit D la plaque homogène dans la figure ci-dessous. Sans faire de calculs d'integrales, en déduire :*

$$\mathcal{I}_1 = \iint_D dxdy; \quad \mathcal{I}_2 = \iint_D xdxdy; \quad \mathcal{I}_3 = \iint_D ydxdy.$$




***Solution :***

$\mathcal{I}_1 = \text{aire}(D) = 7, \quad G = (5/2, 5/2)$ centre de gravité

$\mathcal{I}_2 = \text{aire}(D).x_G = \frac{35}{2}, \qquad \mathcal{I}_3 = \text{aire}(D).y_G = \frac{35}{2}.$

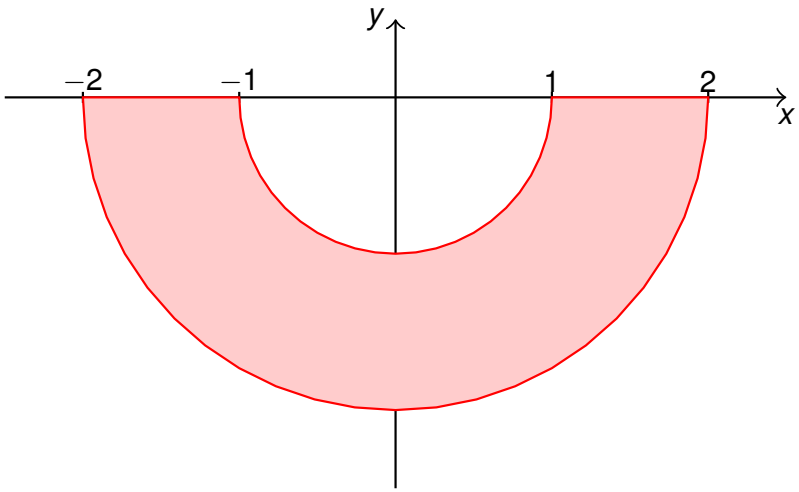
## Exercice 4

*Après avoir représenté graphiquement la région :*

$$D = \left\{(x, y) \in \mathbb{R}^2; \quad y \leq 0, \quad 1 \leq x^2 + y^2 \leq 4\right\}$$

*Calculer les coordonnées de son centre de gravité en supposant la région D homogène.*

***Solution :***




$$x_G = \frac{1}{A}\iint_D x dxdy, \qquad y_G = \frac{1}{A}\iint_D y dxdy$$

où $A = \iint_D 1.dxdy.$

En passant aux coordonnées polaires on a

$$A = \int_{-\pi}^{0} \int_{1}^{2} r dr d\theta = \frac{3\pi}{2}$$

$$\iint_D x dx dy = \int_{-\pi}^{0} \int_{1}^{2} r^2 \cos\theta dr d\theta = 0$$

$$\iint_D y dx dy = \int_{-\pi}^{0} \int_{1}^{2} r^2 \sin\theta dr d\theta = -\frac{14}{3}.$$

On déduit que

$$x_G = 0, \qquad y_G = -\frac{28}{9\pi}.$$